Harvard SEAS — Master of Engineering (ME)

Statement of Purpose: Distributed Systems Architecture and Resilient Infrastructure Engineering

Candidate:	Sham Satish Thakare (PRN: 72159719L, Seat: B191102032)
Target Program:	Master of Engineering (ME), Harvard SEAS
Application Deadline:	December 01, 2026 (5:00 PM EST)
Professional Experience:	Pre-Doctoral Researcher, Capital One Systems Research (06/2024–Present)
Certified Credentials:	OCI Certified Architect, Microsoft SQL AI Developer, Google AI Professional
Target Faculty:	Prof. Minlan Yu, Prof. James Mickens, Prof. Stratos Idreos

Statement of Purpose

As global digital infrastructure scales to exabyte volumes and sub-millisecond latency budgets, software engineering requires a profound synthesis of distributed systems theory, modern hardware-conscious systems programming, and disciplined architectural design. Over the past two years, I have operated at the leading edge of this challenge: conducting pre-doctoral systems research at Capital One Systems Research under the mentorship of Distinguished Engineer Yuri Shelke, while earning 5 industry cloud and AI certifications from Oracle, Microsoft, and Google. At the Harvard John A. Paulson School of Engineering and Applied Sciences (SEAS), I seek admission to the Master of Engineering (ME) program to deepen my mastery of high-throughput networked systems, scalable data platforms, and fault-tolerant distributed consensus engines.

My academic career at Savitribai Phule Pune University provided an exceptional technical and mathematical foundation. Graduating with a Bachelor of Engineering in Artificial Intelligence and Data Science with First Class with Distinction (8.70 CGPA) and achieving Rank 1 in my Final Year (9.53 SGPA), I was awarded the university Gold Medal in Mathematics (2024) for academic excellence. My transcript reflects the highest possible grade (Grade O, 10.0/10.0 Outstanding) across foundational systems disciplines: Distributed Computing (417531), Operating Systems (217521), Cyber Security (317530), and Senior Capstone Project. Beyond academic coursework, I demonstrated institutional leadership as President of the Social Volunteer Club (2023–2024), organizing regional educational outreach and disaster-relief operations that mobilized over 250 volunteers and impacted 1,200+ beneficiaries across Maharashtra.

In my research at Capital One Systems Research, I focused on high-performance infrastructure engineering. To address catastrophic tail latency in multi-region cloud databases, I designed and implemented quorumshift, a modern C++20 Raft consensus engine. Under asymmetric network partitioning, conventional consensus protocols suffer from chronic leader election cycles and buffer bloat. I formulated a dynamic quorum re-weighting algorithm bounded by analytical clock drift (Δ-bounded drift) that dynamically shifts vote weight across healthy availability zones while preserving majority intersection invariants ($Q_i \cap Q_j \neq \emptyset$). Benchmarks demonstrated the complete elimination of p99 latency spikes during partition injection. In tandem, I architected medirush-safeagent, a high-assurance runtime interception platform with zero unhandled side-effects, demonstrating my capability to architect software platforms where failure is not an option.

The Harvard Master of Engineering program represents the definitive bridge for my professional and academic ambitions. The program's rigorous curriculum and project-driven capstone structure offer the perfect arena to advance my systems engineering capability. I am particularly eager to study under Prof. Minlan Yu, whose work in high-speed networked systems, automated traffic management, and cloud telemetry sets the standard for modern networking. I also seek to study distributed fault-tolerant architectures with Prof. James Mickens, exploring robust state-machine replication and resilient mobile/cloud computing. Furthermore, the data systems architecture research of Prof. Stratos Idreos in the Data Systems Laboratory (DASL) directly aligns with my interest in memory-efficient data structures and hardware-accelerated computation.

I will bring to Harvard SEAS a proven track record of academic distinction, validated systems implementation in C++20, and proven institutional leadership. I look forward to contributing actively to the Harvard engineering community.